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**Advanced Cooling Techniques
for High Pressure Hydrocarbon-
Fueled Rocket Engines**

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Abstract

The regenerative cooling limits (maximum chamber pressure) are defined for oxygen/hydrocarbon (Methane, Propane, and RP-1) rocket engines over a thrust range of 20,000 to 600,000 lbf for a reusable life of 250 missions. Chamber pressure limits are first defined without a hot-gas wall carbon layer (unenanced designs). Cooling enhancement chamber pressure limits are then established for seven thermal barriers (carbon layer, ceramic coating, graphite liner, film cooling, zoned combustion, transpiration cooling, and a combination of two of the above). The maximum regenerative-cooled chamber pressure is attained with the oxygen/methane propellant combination.

Introduction

This effort was conducted to investigate advanced cooling techniques for high-chamber-pressure oxygen/hydrocarbon rocket engines. This effort provides NASA and vehicle designers with chamber pressure limits for future hydrocarbon regeneratively cooled thrust chamber assemblies, and the potential of extending the chamber pressure limits when advanced cooling techniques are applied.

This study was initiated by combining sophisticated thermal, structural, cyclic life, and creep-stress rupture analyses into one computerized program to optimize combustion chamber and nozzle designs for reusable engines. These analyses employ the latest empirical relationships and design criteria derived from experiences on the reusable Space Shuttle Main Engine (SSME). The material utilized for this study are identical to those employed on the SSME. These designs also utilize the latest in manufacturing technology developed under the SSME program.

The combustion chambers are configured of bi-width milled passages fabricated of NARloy-Z within currently established manufacturing limits. Cyclic life and reusability of the SSME design have been experimentally verified. NARloy-Z is a Rockwell International (Rocketdyne) material especially developed for the SSME. It is a high-strength, high-conductivity copper-base alloy possessing excellent cyclic life and creep-rupture properties. The nozzles utilize A-286 tubes. Thermal/structural/life analyses are identical to those verified for the SSME combustion chamber and nozzle designs.

Study guidelines are presented in Table 1. Analyses were conducted for both gas-generator cycle and staged-combustion cycle designs. The combustion chamber/nozzle system coolant $\Delta P/P_c$ is 0.60 similar to the SSME. When the maximum guideline chamber pressure for each thrust level was met, the minimum P_{inlet}/P_c of the coolant established the design for

Table 1. Study Guidelines

<u>PROPELLANTS</u>			} MAXIMUM SPECIFIC IMPULSE
O ₂ /RP-1 AT 2.8 MR	}		
O ₂ /C ₃ H ₈ AT 3.1 MR			
O ₂ /CH ₄ AT 3.5 MR			
<u>ENGINE SIZE</u>			
<u>LBF</u>	<u>ε</u>	<u>P_c LIMITS, PSIA</u>	
20,000	400:1 VAC	4000	
150,000	400:1 VAC	5000	
600,000	80:1 SL	6000	
<u>PERFORMANCE</u>			
98% ENERGY RELEASE EFFICIENCY			
<u>DESIGN SAFETY FACTORS</u>			
STRESS = 1.1σ _{YIELD} AND 1.5σ _{ULTIMATE}			
LIFE = 4.0			
<u>ENGINE LIFE</u>			
250 X 4 = 1000 CYCLES			
10 X 4 = 40 HOURS			
<u>COOLANT TEMPERATURE CONDITION/LIMITS, R</u>			
	<u>INLET</u>	<u>COKING (T_{wc})</u>	
RP-1	530	1060	
C ₃ H ₈	420	1320	
CH ₄	210	1760	
<u>ENGINE SYSTEM</u>			
GAS GENERATOR, P _{IN} /P _c = 1.8			} ΔP/P _c = 0.60
STAGED COMBUSTION, P _{IN} /P _c = 2.25			

the limiting chamber pressure. This criterion was disregarded for the film-cooling-enhanced designs to determine extended chamber pressure versus thrust trends. Other enhancement techniques were within the P_c guideline limits, and do not represent the maximum regeneratively cooled chamber pressure when a reduced P_{inlet}/P_c was used.

All combustion chamber and nozzle designs were detailed to the extent of configuration, coolant passage geometry, and local thermal and life characteristics. All designs, unenhanced and enhanced, were optimized to attain the maximum chamber pressure for the specified life (250 missions x a safety factor of 4) for 1000 cycles.

Design Considerations

Combustion Chamber

Combustion chamber geometry (liner hot-gas wall contour) affects the weight, performance, heat transfer, and life. Influencing parameters are length, contraction area ratio, and wall contour. The optimum combustion chamber design is one with minimum engine weight, minimum hot-gas wall temperature (maximum life), and maximum performance. In addition, engine constraints are imposed on the chamber design as related to available coolant pressure drop and flowrate requirements established for an engine

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balance with minimum oxidizer and fuel pump discharge pressure. The optimum coolant passage geometry for maximum life (minimum T_{wg}) normally approaches the minimum wall thickness and minimum channel and land widths. These values are dictated by manufacturing limits. This study used minimum channel width and land width of 0.030 inch, with minimum wall thickness of 0.025 inch. A maximum limit of 4:1 on channel height/width was also recognized.

Thrust chamber nozzle contour primarily affects the performance and weight of an engine. Location of the nozzle attachment to the combustion chamber is of great significance, affecting life and engine weight. The optimum attach point is located as near to the throat as possible to still meet nozzle tube life and coolant ΔP . Care must be taken to ensure the chamber throat coolant is at an efficient bulk temperature. The optimum attachment location normally is between an area ratio of 5:1 and 10:1. The higher the chamber pressure, the larger the attachment area ratio. The minimum tube wall thickness normally provides maximum life for minimum coolant pressure drop. A minimum tube wall thickness of 0.008 inch was utilized. These design considerations are typical of those employed to attain the maximum cooling and life for the SSME (Ref. 1).

The performance of a liquid-propellant rocket engine system is dependant on how efficiently the propellants can be injected, combusted, and then expanded supersonically through a nozzle. The combustion efficiency (η_{c*}) depends on injector design and combustor contour (length and contraction ratio). Additional thrust is obtained through optimum expansion of these combusted gases as related to nozzle thrust coefficient efficiency (η_{c_F}). Therefore, certain design criteria must be met to attain realistically high performance. The injector/chamber geometry of this study produces a 98% energy release efficiency (η_{c*}). Nozzles are typically designed for an expansion efficiency (η_{c_F}) of 97.5%. These efficiencies are related to the theoretical optimum efficiencies attainable.

The combustion chamber injector-to-throat length and the contraction ratio are shown in Fig. 1 and 2

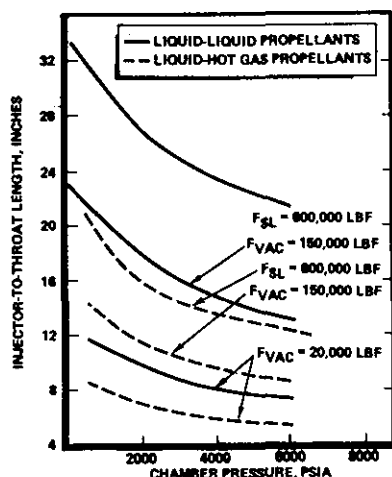


Fig. 1. Injector-to-Throat Length Parametric Data

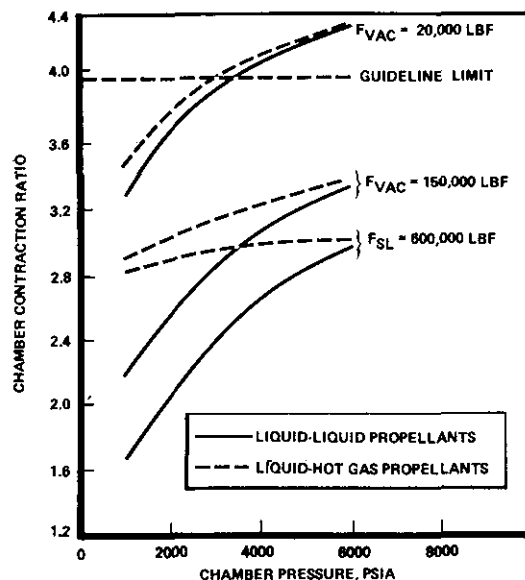


Fig. 2. Chamber Contraction Ratio Parametric Data

for the three thrust levels evaluated using both liquid-liquid and liquid-gas propellant injection. These curves are based on empirical relationships obtained from numerous engine programs utilizing various propellant combinations and injector designs.

The oxidizer (oxygen) is not truly a liquid at the point of injection in that it has a high vapor pressure, low viscosity, and virtually no surface tension. The liquid fuel is the limiting propellant with atomization, vaporization, and mixing being the limiting mechanisms. Liquid-liquid engines require longer residence time (longer length) for complete mixing and combusting of the propellants than do liquid-hot-gas engines. The injector-to-throat length is increased as the thrust is increased, primarily because of the coarser injection pattern as the thrust is increased. The chamber length can be decreased with increasing chamber pressure because reaction rates increase with chamber pressure.

Higher chamber pressure operation also leads to larger injection orifices since the mass flowrate per unit injector face area is increased with increasing chamber pressure. Therefore, an increase in contraction ratio is desirable as the chamber pressure increases (Fig. 2).

The staged-combustion cycle analyses utilized the gas-gas injection design criteria. The propane and methane fuels were treated as hot gases for the gas generator cycle because of the large temperature increase each incurs during cooling. The liquid-liquid curves in Fig. 1 and 2 were initially used for the gas generator cycle design with RP-1 as the fuel using conventional type injectors. However, it is believed that low thrust per element type injectors can be developed to attain an energy release efficiency of 98%. Oxygen/RP-1 design analyses were also conducted for these prospective gas generator cycle designs for comparison.

Selection of combustion chamber length and contraction area ratio is primarily dictated by heat

transfer, system weight, and combustion performance. The combustion chamber coolant pressure drop is primarily a function of the chamber length and contraction area ratio.

These designs use a cylindrical section at the injector end and a large radius to convergence that provides injector-to-combustor wall compatibility. A generous upstream convergence radius joining the cylindrical combustion chamber section with the throat entrance ramp provides a smooth transition to avoid flow separation at the turn or impingement at the initial portion of the chamber convergence ramp on the hot-gas wall. It also prevents degradation of the coolant film coefficient due to reverse curvature effects.

Steep throat entrance angles result in late boundary-layer attachment and consequent high throat heat flux; whereas, shallow entrance angles (< 30 degrees) result in continuous boundary-layer growth with minimum throat heat flux. Shallow entrance angles do, however, impose a slightly greater integrated heat load due to a lower average contraction area ratio. A 25- to 30-degree convergence angle was selected for these designs favoring lower throat heat flux for maximum life.

The radius of curvature relative to the throat radius (r_c/r_t) entering the throat transonic region affects both the distortion and location of the sonic line and the distribution of the transonic flowfield in the throat region. The large value of the radius ratio used for these designs (r_c/r_t) = 1.0, provides a minimum distortion and supersonic flow divergence loss.

Life Definition and Analysis Logic

Life is the number of duty cycles a nozzle or main combustion chamber can achieve prior to crack initiation. The main chamber liner and nozzle are designed for 250 duty cycles, for an accumulated run duration of 10 hours. A safety factor of four was utilized for the life analysis. This means the predicted life must equal or exceed 1000 duty cycles for an accumulated operating time of 40 hours. Life of the regeneratively cooled nozzle and main combustor liner is determined by the accumulation of low-cycle fatigue and creep rupture damage, as shown in Fig. 3.

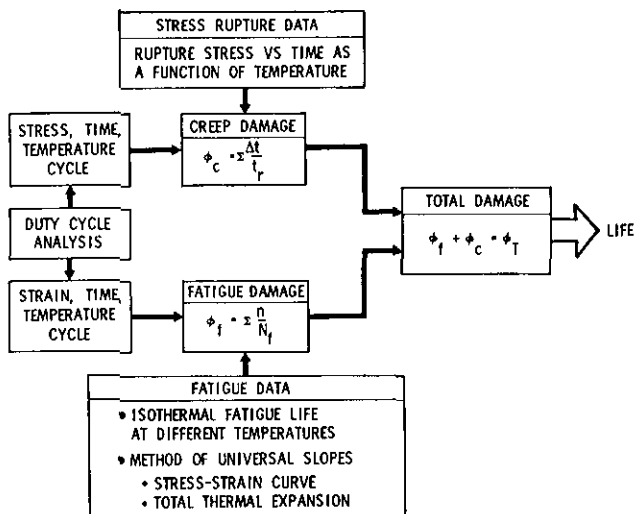


Fig. 3. Life Analysis Logic

Creep Rupture

Creep-rupture damage is a function of time at operating stress and temperature levels. Creep-rupture damage is evaluated using the "life fraction rule" which is the summation of time at stress divided by the time to rupture at these operating stress and temperature levels:

$$\phi_c = \sum \left(\frac{t}{t_r} \right)_{\sigma} \quad (1)$$

Low-Cycle Fatigue

Low-cycle fatigue damage is a result of thermal and pressure cycling. Low-cycle fatigue is a function of the hot-gas wall total effective strain, which is primarily cyclic plastic strain. The relationship between the effective cyclic strain and fatigue life (N_f) of materials has been established by the form:

$$N_f = \left(\frac{C}{\epsilon_{eff}} \right)^a \quad (2)$$

where C is a function of material characteristics and properties, i.e., fracture ductility or reduction in area (RA). The exponent (a) has been shown experimentally to be sensitive to temperature, environmental effects, and fracture mode (transgranular and intergranular).

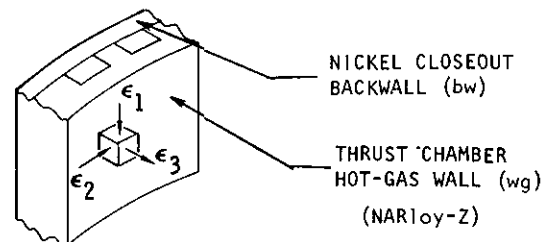
The total effective strain is comprised of:

1. Thermal strain due to the temperature differential between the liner hot-gas wall and the structural jacket
2. Strains caused by the hydraulic pressure stresses attributed to the coolant to hot-gas pressure differential
3. External axial strains from nozzle forces imparted to the structural chamber jacket (Inconel 718) and transferred to the NARloy-Z liner

The thermal strain represents approximately 90% of the total effective strain because of the large temperature gradient between the hot-gas wall and backwall structure. The maximum strain occurs on the hot-gas wall where the maximum temperature gradient between the hot-gas wall and backwall structure ($T_{wg} - T_{bw}$) exists. The total effective cyclic strain (ϵ_{eff}) as derived from the Von Mises criteria, is

$$\epsilon_{eff} = \frac{\sqrt{2}}{3} [(\epsilon_1 - \epsilon_2)^2 + (\epsilon_2 - \epsilon_3)^2 + (\epsilon_3 - \epsilon_1)^2]^{0.5} \quad (3)$$

for an element of material (ϵ_1 , ϵ_2 , and ϵ_3):



Based on the "constant volume assumption" of plasticity theory

$$\epsilon_3 = -\epsilon_1 - \epsilon_2 \quad (4)$$

the effective biaxial cyclic strain of the main combustor then becomes

$$\epsilon_{\text{eff}} = \frac{2}{\sqrt{3}} [(\epsilon_1)^2 + (\epsilon_1 \epsilon_2) + (\epsilon_2)^2]^{0.5} \quad (5)$$

Thermal Strains. The thermal strains on the combustor liner hot-gas wall are attributed to the temperature gradient between the liner hot-gas wall and the backwall (structural jacket).

$$\epsilon_{\text{thermal (uniaxial)}} = [\alpha(T_{\text{wg}} - T_{\text{bw}})] \quad (6)$$

The maximum biaxial strain is located on the hot-gas wall and is compressive during mainstage and tensile during the start transient since the coolant liner is restrained axially and tangentially by the structural jacket.

For channel wall construction

$$\epsilon_1 = \epsilon_2 \quad (7)$$

Equation 5 then simplifies to:

$$\epsilon_{\text{eff}} = (2 \epsilon_1)^{\beta} = [2 \alpha(T_{\text{wg}} - T_{\text{bw}})]^{\beta} \quad (8)$$

for a biaxial strain condition, where β is a correction factor to correlate hot-fire test results with finite element strain analyses.

For the nozzle tubes, the equivalent biaxial strain is determined by Eq. 5 where ϵ_2 is the tangential strain across the tube wall which is unrestrained

$$\epsilon_2 = \frac{\alpha(T_{\text{wg}} - T_{\text{bw}})}{2} \quad (9)$$

The steady-state pressure stresses determine the creep damage and are determined by

$$\sigma_B = \left[\frac{P_{\text{coolant}} - P_{\text{gas wall}}}{2} \left(\frac{W}{t} \right)^2 \right] \quad (10)$$

for channel walls, and

$$\sigma_S \left[\frac{P_{\text{coolant}} - P_{\text{gas wall}}}{2} \frac{R}{t} \right] \quad (11)$$

for tube walls, where W = channel width, t = wall thickness, and R = tube wall radius.

Materials Selection

The materials selected for the combustion chambers and nozzles are discussed below.

Combustion Chamber

A high thermal conductivity material is required for minimum wall temperature and minimum thermal strain. Maximum cyclic life demands a high-ductility material. Therefore, material requisites of high thermal conductivity and high ductility necessitates

the selection of a compatible copper-base alloy. The strength of the material and heat transfer aspects dictate the coolant channel geometry, which influences the coolant pressure drop, engine weight, life, manufacturing requirements, and cost. It is desirable to have a high-strength material. NARloy-Z a zirconia-silver-copper alloy, was developed by Rocketdyne for the SSME main combustion chamber liner. It exceeds the life requirements, and possesses a relatively higher strength than other commercially available copper base alloys.

Nozzle

Wrought A-286 was selected as the nozzle material. This material was also selected for the SSME nozzle. It is a moderate strength, heat resistant alloy, a high-strength alloyed stainless steel hardenable by solution treating and aging. The alloy is always used in its hardened condition. The material is entirely satisfactory for use at all cryogenic temperatures down to 37 R with retention of excellent toughness and ductility properties. At elevated temperatures, the alloy retains moderate strength up to 1660 R. The alloy is readily formable in the annealed or solution-treated condition.

Load-Carrying Structure

A high-strength super alloy (Inconel 718) was used as the load-carrying structure for the chamber and nozzle designs. The backup structural loads were considered to be within the yield strength ($1.1 \sigma_y$) of the Inconel 718 structure and only imparted these loads to the regenerative cooled chamber and nozzle circuits.

Combustion Gas Heat Transfer

A Rocketdyne boundary layer analysis was used to calculate the hot-gas heat transfer coefficient profiles for the chamber and nozzle designs. This program utilizes an integral method to solve the momentum and energy equations. Solution of these equations is accomplished using semi-empirical relations between the Stanton number and the energy thickness, and between the skin friction coefficient and momentum thickness. The resulting equation for turbulent flow is of the form:

$$N_{ST_{\infty}} = \frac{0.0122}{(Re_{\phi})^{0.25}} \left(\frac{\rho_{\infty}}{\rho_r} \right)^{1/2} \left(\frac{\mu_r}{\mu_{\infty}} \right)^{1/4} \frac{1}{N_{Pr}^{2/3}} \quad (12)$$

The convective film coefficient (h_g) is then found from the relation:

$$h_g = \rho_{\infty} U_{\infty} C_P N_{ST_{\infty}} \quad (13)$$

The Eckert reference temperature is used to evaluate the film properties.

The injector end heat transfer coefficient for any given case is determined by scaling calorimeter chamber test data from the SSME 40K subscale chamber test program (Ref. 2) using flowrate and property corrections based on the standard Nusselt Number correlation:

$$h_{g2} = h_{g1} \frac{k_2}{k_1} \left(\frac{G_2}{G_1} \times \frac{\mu_1}{\mu_2} \right)^{0.8} \left(\frac{Pr_2}{Pr_1} \right)^{0.4} \quad (14)$$

where subscript 1 = reference conditions (40K sub-scale test data) and subscript 2 = O_2 /hydrocarbon conditions. This injector end heat transfer coefficient is then blended into the axial profile derived from the boundary layer program. An example of this is shown in Fig. 4.

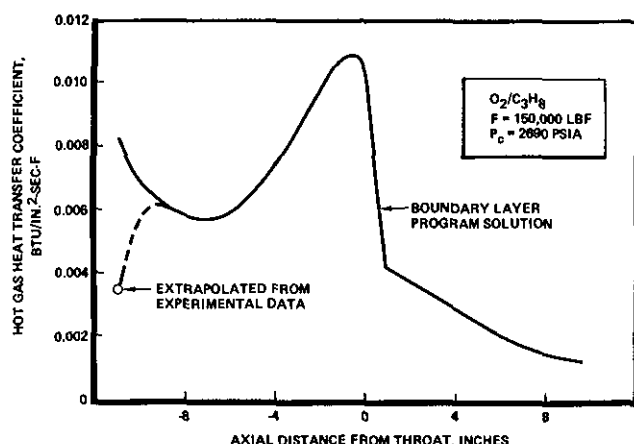


Fig. 4. Hot-Gas Heat Transfer Coefficient Profile

Cooling Correlations

The following cooling correlations were used for this study. These correlations were experimentally determined at Rocketdyne using electrically heated tubes.

$$\text{RP-1 } h_c = 0.0056 (k/D_h) \text{Re}^{0.95} \text{Pr}^{0.4} \phi_R \phi_C \quad (15)$$

where ϕ_R is the roughness enhancement factor and is determined by taking the ratio of the ϵ/D_h of the passage to friction factor for an ϵ/D_h of 0.0001, and ϕ_C is the curvature enhancement factor.

The standard Nusselt number correlation with the roughness enhancement factor included was used for methane cooling:

$$\text{Methane } h_c = 0.023 (k/D_h) \text{Re}^{0.8} \text{Pr}^{0.4} \phi_R \phi_C \quad (16)$$

and

$$\text{Propane } h_c = 0.00696 (k/D_h) \text{Re}^{0.88} \text{Pr}^{1/2} (T_c/T_{wc})^{1.0} \phi_R \phi_C \quad (17)$$

The coolant curvature enhancement factor (ϕ_C) profile utilized for these designs as a function of turning angle is the same profile experimentally determined for the SSME main combustion chamber.

Influence of Thrust Level

The low thrust (20,000 lbf) and intermediate thrust (150,000 lbf) designs have similar optimization requirements. Because of the small amount of fuel available for cooling relative to the hot-gas surface area, a series circuit using all of the fuel is utilized to minimize the coolant temperature rise in the combustion chamber. The combustion chamber is cooled first in an uppass, then the nozzle is cooled in a downpass.

The channel geometry of the low thrust designs are at the guideline minimum limits, while the intermediate thrust designs are slightly above the minimum. Because of the low coolant flowrate and small-diameter nozzle tubes, a high chamber-to-nozzle attachment area ratio must be used to keep the nozzle coolant velocity high enough relative to the local heat flux to meet the cycle life requirement. The minimum attachment area ratio that meets the above criteria is 10:1 for the low thrust and intermediate thrust designs.

At the high thrust level (600,000 lbf), there is so much coolant flow available that only a fraction of the fuel can be used for cooling the chamber while maintaining reasonable channel dimensions. A split flow (uppass chamber, downpass nozzle) cooling circuit is required with approximately 50% of the available coolant going to the chamber and the remaining 50% to the nozzle. The channel width must be optimized by trading off the increased flowrate with the decreased cooling efficiency as the channel width is increased. A throat region and combustion zone channel width of approximately 0.055 inch was determined to be optimum. The chamber-to-nozzle attachment area ratio is decreased to 5:1.

Influence of Engine Cycle

The staged-combustion (SC) cycle operates at a coolant $P_{inlet}/P_c = 2.25$, while the gas generator (GG) cycle uses a coolant $P_{inlet}/P_c = 1.8$. This requires an adjustment to channel and tube wall thickness to accommodate the higher pressure gradient across the wall for staged combustion designs. The design results show that the chamber pressure limits of the two cycles considered are within 3% at all thrust levels.

Influence of Propellant Combination

The unenhanced hot-gas convective heat transfer is similar for all three propellant combinations. Therefore, their chamber pressure limits, and coolant passage geometry are derived from the fuel cooling capabilities. However, for the O_2 /RP-1 propellant combination, the RP-1 decomposition temperature is very low. Because of this low coking temperature limit, the chamber pressure regenerative cooling limits are quite low.

Design Results

The results of the unenhanced regeneratively cooled limit designs are presented in Fig. 5.

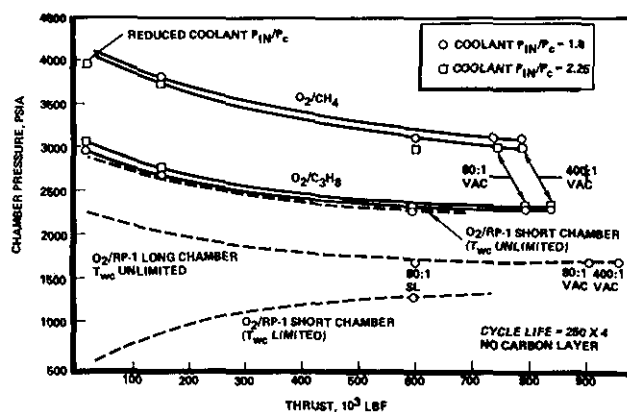


Fig. 5. Effect of Thrust on Regenerative Cooling Limit (Unenhanced Designs)

1. $O_2/RP-1$ design cooling limits are at very low chamber pressures as dictated by a limiting $RP-1$ coolant decomposition temperature at $T_{wc} \leq 1060$ R. The $O_2/RP-1$ cooling limit chamber pressure decreases with a decrease in thrust. This is because the propellant flowrate and, therefore, the coolant flowrate to total head load and surface area ($\dot{w}_c/\dot{Q}$) decreases with a decrease in thrust level. These designs are not life-limited.
2. The cooling limit chamber pressure of the O_2/CH_4 and O_2/C_3H_8 designs decreases with an increase in thrust. These designs are life-limited, but are not limited by a decomposition wall temperature. The life-limited $O_2/RP-1$, O_2/CH_4 , and O_2/C_3H_8 designs result in an increased chamber pressure with a decreased thrust level. This is not typical since the ($\dot{w}_c/\dot{Q}$) trend favors the higher thrust level for cooling, as reflected for the T_{wc} limited $O_2/RP-1$ designs. One basic reason the O_2/CH_4 and O_2/C_3H_8 designs reflect reverse trends is related to a constant low cycle fatigue life criterion. These curves do not reflect a constant coolant wall or hot-gas wall temperature.

For a constant life, the low thrust designs operate at higher T_{wg} because of the higher bulk temperature in the high heat flux regions. Cyclic life is a function of the temperature gradient between the hot-gas wall (T_{wg}) and the closeout structure back-wall temperature (T_{bw}), which is approximately the bulk temperature (T_b). Therefore, cyclic life (4×250) is met for all thrust levels but requires lower T_{wg} for the higher thrust designs because of the lower coolant temperature in the throat and convergence region of the chamber. Another contributor for the reversed P_c versus F trend is the combustion chamber length (injector to throat) and contraction area ratio criteria to attain a 98% energy release efficiency. These criteria result in a combustor (injector to throat) average heat flux ($\dot{Q}/A$) decrease for the lower thrust designs.

Enhanced Designs (Thermal Barriers)

Three physical thermal barriers and three fluid thermal barriers were evaluated plus a combination of two of the heat barriers.

The physical barriers considered for cooling enhancement were carbon-layer, ceramic coating (ZrO_2), and ATJ graphite liner. These are applied to the parent material hot-gas-wall surface as thermal resistances. These thermal barriers were not applied to the tubular nozzle, except as required for some $O_2/RP-1$ nozzles to maintain a $T_{wc} \leq 600$ F. These thermal barriers were utilized for the channel constructed combustion chamber.

The fluid thermal barriers considered for cooling enhancement were film cooling, transpiration cooling, and zoned combustion. All three fluid barriers were allotted a 2% performance loss.

Carbon Layer

A carbon coating was selected as the renewable physical thermal barrier. The mechanism of deposition of the carbon layer on the wall is not fully understood. The elemental carbon is formed by cracking of the fuel. Evidence exists that shows the carbon layer builds up during the first few milliseconds of operation and then undergoes periodic spalling due to thermal and dynamic stresses until an equilibrium value is achieved.

A number of factors appear to influence the amount of carbon layer buildup: chamber pressure, mixture ratio, type of fuel, injector design, amount of film cooling, and wall temperature. Reference 3 contains combustion chamber heat load data for low chamber pressure FLOX/methane, FLOX/propane, and FLOX/butene-1 propellants over a range of mixture ratios. These data were reviewed to determine the effect of fuel type and mixture ratio on the carbon layer. It was found that the carbon layer resistance correlated best with the carbon atom fraction in the combustion gas, as shown in Fig. 6. By correlating with the carbon atom fraction the carbon layer resistance can be determined for any hydrocarbon fuel at any mixture ratio.

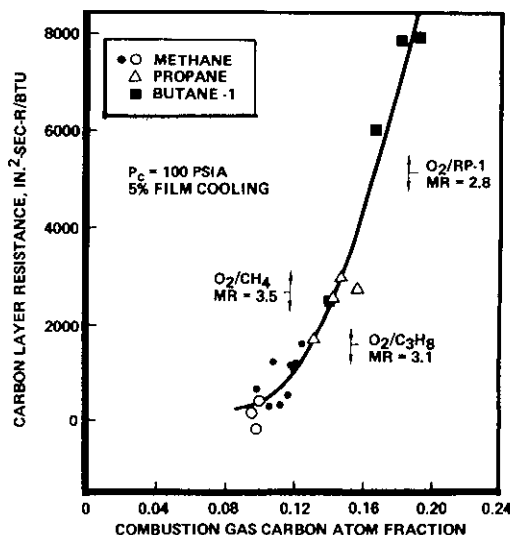


Fig. 6. Effect of Fuel on Carbon Layer Resistance

These tests were all run at a chamber pressure of approximately 100 psia. Rocketdyne developed a correlation for carbon layer resistance as a function of the combustion gas mass velocity (Ref. 4) for high chamber pressure. This correlation is shown in Fig. 7 (the dashed line). The test data in Fig. 7 are for $O_2/RP-1$ propellants. Additionally, a large amount of film cooling ($> 10\%$) was used during the tests. The carbon layer resistance for each of the three propellant combinations at their respective mixture ratios is taken from Fig. 6 and plotted in Fig. 7 for lower mass velocities. Lines are then extended from these three points parallel to the Rocketdyne-developed curve for higher mass velocities.

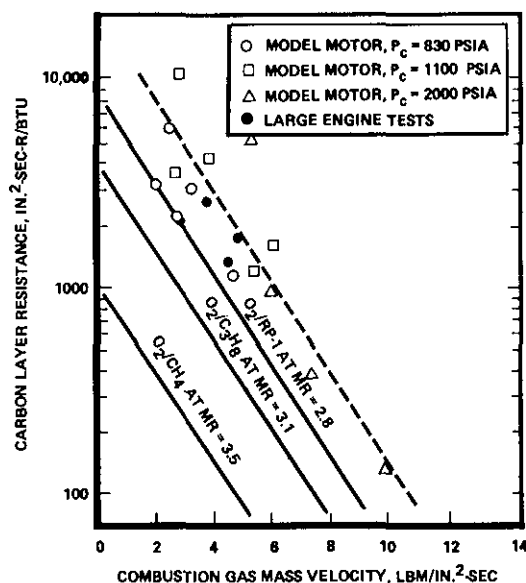


Fig. 7. Carbon Layer Resistance Correlations

There is a considerable amount of uncertainty in these equations, particularly when they are applied at high chamber pressures. However, until additional data become available, the solid lines of Fig. 7 represent the best available measure of the effectiveness of carbon layers. As shown in Fig. 7, the carbon layer thermal resistance (x/k) is greatly dependent on hot-gas mass velocity (G_g) and propellant combination. This means the carbon layer thermal resistance is large in the combustion chamber and supersonic regions compared to the higher heat flux throat region. Therefore, the enhancement benefit to be attained requires enlarging the coolant passages in the regions upstream and downstream of the throat to provide a higher coolant mass velocity in the throat region. This tradeoff requires adjusting the unenhanced designs to provide greater cooling of the throat region.

Ceramic Coating

Zirconium oxide (ZrO_2) was selected as the ceramic coating for a physical thermal barrier. Zirconium oxide is a low thermal conductivity material that provides an excellent thermal barrier. However, because of that, and the fact that it is a low-ductility material, the use of its application is limited to thin thicknesses for high chamber pressure. To maintain the surface temperature limit (< 3500 R), the thicknesses employed were between 0.001 and 0.005 inch. The large thermal gradients across the coating thickness means large thermal strains. This also dictates the surface temperature limit. It is the most utilized ceramic coating for reducing hot-gas wall temperatures in high heat flux rocket engines.

The utilization of ZrO_2 coatings enhances the life of the parent combustor NARloy-Z material. The cyclic life of the ZrO_2 coating is undefinable at this time. It is most likely that the ceramic coating will have to be replenished a number of times during the life of the NARloy-Z chamber. The chamber designs utilized a bi-thickness approach (thinner in the throat region) for maximum chamber pressure limits.

Graphite Liner

The graphite liner analysis guidelines utilized ATJ graphite at a maximum surface temperature of 2500 F, and a manufacturers wall thickness of 0.100 to 0.500 inch. The graphite wall thickness limits were a function of chamber diameter. The thicker liner was for the larger thrust engines. A comparative assessment showed that the minimum ATJ graphite liner thickness specified by the manufacturer and the maximum guideline surface temperature were incompatible for these high heat flux-high chamber pressure designs. The ATJ graphite wall thickness required to meet the 2500 F surface temperature was between 0.015 and 0.030 inch. Therefore, other graphite liner prospects were reviewed, i.e., pyrolytic graphite and "graphoil-laminate" graphite. These types of graphites can be applied in thinner layers (> 5 mils). However, the thermal conductivity perpendicular to the hot wall results in thermal conductances greater than an order of magnitude beyond those of ATJ graphite, being near the thermal resistance of ZrO_2 . Because of the decreased conductivities of pyrolytic graphic materials, their liner thickness must be on the order of several thousandths of an inch thick for high chamber pressures, which is outside of the manufacturing limits. In summary, graphite liners are only applicable for space engine type low heat flux combustion chambers.

Fluid Thermal Barriers

Film Cooling

Film cooling was selected as one of the three thermal fluid barriers. It has been used to some degree in most liquid propellant rocket engines. A number of film-cooling heat transfer models was evaluated. Most of the existing analytical models assume that the presence of the film does not affect the heat transfer coefficient between the wall and film, and concentrate on prediction of the effective film temperature as a function of distance from the injection point. These models utilize a "dry wall" heat transfer coefficient and assume no mixing of the free stream main core flow with the film cooling.

A more accurate approach to compute the heat transfer rate between a gaseous film coolant and wall is to predict both a heat transfer coefficient and an effective film temperature. This is particularly true when the film-coolant properties are different from the free-stream gas. One such correlation for predicting the film heat transfer coefficient was developed at Rocketdyne (Ref. 5), and assumes equal Mach numbers of film and free-stream gases to relate the heat transfer coefficient of the film to the dry wall heat transfer coefficient as

$$\frac{h_f}{h_g} = \frac{C_{p_f}}{C_{p_g}} \left(\frac{P_r}{P_{r_f}} \right)^{2/3} \left[\frac{\gamma_g M_g T_g}{\gamma_f M_f T_f} \right]^{1/2} \quad (18)$$

This approach accounts for a mass balance in the film where the concentration difference is expressed as a ratio of flowrates and injection mass fluxes:

$$\frac{\partial}{\partial x} \frac{w_f}{p} = G \frac{\partial \Delta}{\partial x} + Le \frac{h_g}{C_{p_g}} \frac{m_f w_c}{m_c w_f} \quad (19)$$

The entrainment of the free-stream gas into the film is accountable by impingement (the first term) and by diffusion (the second term).

The film cooling efficiency used was 20%. This is compatible with design considerations and efficiencies of other experimenters, accounting for molecular weight sensitivity and slot injection characteristics.

Transpiration Cooling

Transpiration cooling is applied by removing a regenerative-cooled section of the throat and by replacing it with a section that inputs some fuel coolant through slots or a porous material. The optimum type of transpiration cooling is a wafer-type which also benefits from convective cooling and conduction as the transpiration coolant is injected through slots to the hot-gas wall. However, this study considers only the conventional type of transpiration cooling. It is assumed that the coolant is injected through a porous-type material, and there are no convective or conduction benefits. Also, only CH_4 and C_2H_6 are considered because of the potential coking problems with RP-1 fuel.

Considerable analytical and experimental effort has been expended in past years in the area of transpiration cooling. Most analyses, however, considered laminar flow systems, largely because this type is more amenable to solution. There has been some work in the area of turbulent flow.

The successful turbulent correlation of the Rannie equation for similar gas injection (Ref. 6) suggested that his approach has a reasonably firm basis for extension to foreign gas injection. The original equation derived by Rannie is:

$$\frac{T_{aw} - T_c}{T_{wg} - T_c} = \left[1 + \frac{C_{p_c}}{C_{p_g}} \left(1.18 \text{Re}_b^0 - 1 \right) \left(\frac{-37 \frac{G_c}{G_g} \text{Re}_b^{0.1}}{1 - e} \right) \right] e^{-37 \frac{G_c}{G_g} \text{Re}_b^{0.1} \text{Pr}} \quad (20)$$

where the ratio C_{p_c}/C_{p_g} reflects one of the properties differences between the injected coolant and primary gases. A more accurate analysis would use enthalpy rather than temperature differences.

Calculations for hydrogen injected into air were performed using the preceding analytical technique (Ref. 7). These results are compared in Fig. 8 with the Rannie equation and the simplified heat balance equation, where

$$\frac{T_{aw}}{T_{wg}} = 1 + \frac{1}{\text{St}} \frac{C_{p_c}}{C_{p_g}} \frac{G_c}{G_g} \quad (21)$$

which uses the standard relation for Stanton number:

$$\text{St}_x = \left(\frac{h_g}{G_g C_{p_g}} \right)_x \quad (22)$$

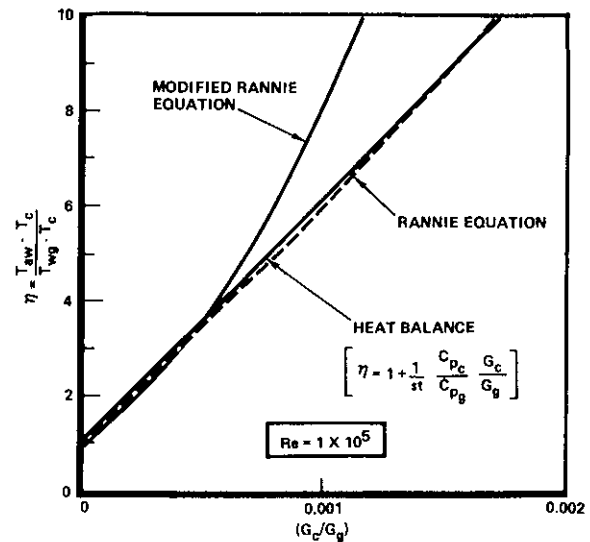


Fig. 8. Comparison of the Modified Analysis for Foreign Gas Injection With the Rannie Equation for Hydrogen Injection Into Air

It is shown that the Rannie relation and the heat balance relation are nearly identical. It is known, however, that the effect of "blowing" is to reduce the Stanton number and, thereby, the coolant flow-rate requirements so that the modified analysis would appear more appropriate. However, because of the simplicity of the heat balance relationship, it was used in this study for the transpiration cooling analyses where the transpiration cooled area is:

$$A_{TC} = \frac{\dot{w}_{TC} C_{p_c}}{\left(\frac{T_{aw} - T_c}{T_{wg} - T_c} \right) - 1 h_{g_x}} \quad (23)$$

Zoned Combustion

Zoned combustion (mixture ratio bias) is a means of normally reducing the wall heat flux by running the outer perimeter of injector elements at a mixture ratio below nominal (where the nominal mixture ratio is below stoichiometric), while running the core elements at a mixture ratio above nominal. In theory, this provides a reduced temperature combustion gas along the wall with a high-temperature core flow. From a cooling standpoint, the most effective case of zoned combustion would be the limiting case of film cooling where the outer zone mixture ratio, in effect, becomes zero.

The zoned combustion approach is graphically presented in Fig. 9 for the three thrust levels for the O_2/CH_4 propellant combination. The guideline 3% performance loss is shown by a 3% vertical displacement from the peak performance (nominal mixture ratio). The outer zone and core mixture ratio for a constant injection pV is a straight line intersecting the 3% vertical offset. The slope of this line is determined by the percentage of outer zone mass flowrate to the total mass flowrate, as depicted for the three thrust levels. The thermal analysis utilized the convective heat transfer coefficient and adiabatic wall temperature associated with the reduced zone mixture ratio. In general,

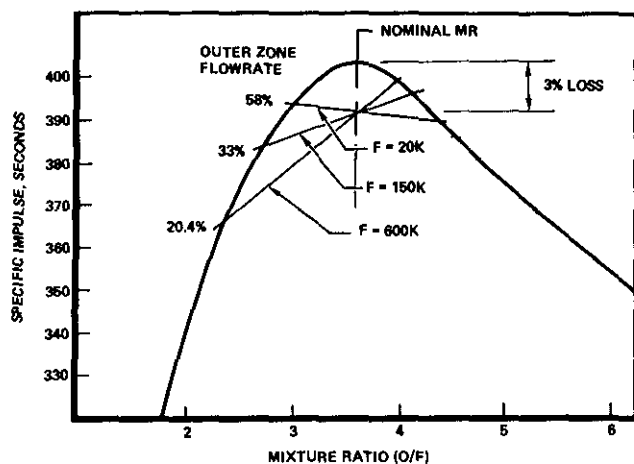


Fig. 9. Zoned Combustion Graphical Approach for the Three Thrust Levels for O_2/CH_4 Propellant Combination

it was handled as a low mixture ratio case. Undoubtedly, one would expect a reduced heat flux for a lower mixture ratio as denoted by zoned combustion. However, as noted, the outer zone operates at an $\eta_{c*} = 98\%$ as designed for a constant injection ρV . Therefore, the analysis is representative of a chamber designed to operate at this mixture ratio as though it were a constant overall mixture ratio. A theoretical heat flux plateau exists for all propellant combinations for the mixture ratio dome region of maximum energy release efficiency. As noted in Fig. 10, zoned combustion only benefits the 600,000 lbf thrust O_2/CH_4 and $O_2/RP-1$ designs, which represents approximately 10% chamber pressure increase for 3% performance loss.

Combined Thermal Barriers

There were several combinations considered. Transpiration cooling and film cooling were the contenders as fluid barriers, each with a 3% performance loss. Ceramic (ZrO_2) coating and carbon layer

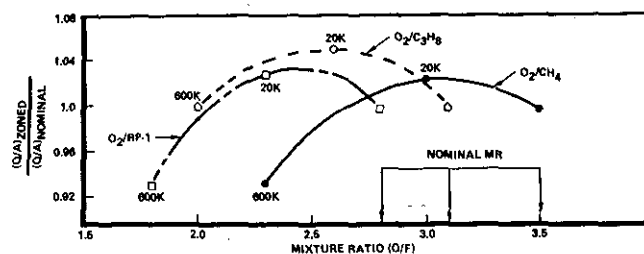


Fig. 10. Comparison of Zoned Combustion Chamber Heat Flux to Unenhanced Chamber Heat Flux at Nominal Mixture Ratio

were the physical barrier contenders. The combination of two fluid barriers was not considered because of the large performance loss. Although transpiration cooling showed the greatest chamber pressure increase potential, it was not selected because of inherent design and fabrication problems and extended development required to make it operational.

ZrO_2 ceramic coating plus film cooling was selected as the combined thermal barrier to provide maximum benefits for increased chamber pressure of the O_2/CH_4 and O_2/C_3H_8 designs.

A carbon layer thermal barrier is extremely overpowering for the $O_2/RP-1$ propellant combination. The next most efficient thermal barrier is a ZrO_2 ceramic coating, as defined for the other two propellant combinations. Therefore, a carbon layer plus ZrO_2 ceramic coating was selected for the combined $O_2/RP-1$ thermal barrier. This combination represents the best thermal barrier for the $O_2/RP-1$ designs. This combination has a great benefit over the combined thermal barriers of the other propellant combinations since it represents no performance loss. The $O_2/RP-1$ barriers are both physical thermal barriers.

Enhanced Design Comparison

A summary of the regenerative cooling limits is presented in Table 2 for the unenhanced and enhanced designs. As noted, many of the enhanced designs are

Table 2. High-Pressure Hydrocarbon-Fueled Engine Regenerative Cooling Limits (Summary)

Propellant Combination	Thrust, lbf	Unenhanced		Carbon Layer		ZrO_2 Ceramic Coating		Film Cooling		Transpiration Cooling		Zoned Combustion		"Other" Combined Thermal Barriers	
		Coolant P_{in}/P_c	Max P_c , psia	Coolant P_{in}/P_c	P_c , psia	Coolant P_{in}/P_c	P_c , psia	Coolant P_{in}/P_c	P_c , psia	Coolant P_{in}/P_c	P_c , psia	Coolant P_{in}/P_c	P_c , psia	Graphite Liner	Combined Thermal Barriers
O_2/C_3H_8	20,000	GG 1.8 SC 2.25	3000 3100	1.8	3150	1.64	4000	1.8	4400	1.63	4000				
	150,000	GG 1.8 SC 2.25	2690 2750											N/A	ZrO_2 + Film Cooling
	600,000	GG 1.8 SC 2.25	2300 2340	1.8	2750	1.8	4000	1.8	3100	1.8	5500				
O_2/CH_4	20,000	GG 1.8 SC 2.25	4100 4050	1.8	4150										
	150,000	GG 1.8 SC 2.25	3800 3720	1.8	4000	1.67	5000	1.8	5400	1.5	5000			N/A	ZrO_2 + Film Cooling
	600,000	GG 1.8 SC 2.25	3110 3000	1.8	3175	1.8	4850	1.8	4200	1.5	6000	1.8	3400		
$O_2/RP-1$	Long Chamber	GG 1.8 SC 2.25	1000 (Ref. Case)	1.8	2000										
	Short Chamber	GG 1.8 SC 2.25	1300	1.8	3300	1.8	2200	1.8	2200		N/A			N/A	ZrO_2 + Carbon Layer
	Long Chamber	GG 1.8 SC 2.25	1710*	1.8	1710*										
	Short Chamber	GG 1.8	2300*												

*Did not meet T_{wc} limit

N/A - Not applicable or no benefits attainable

at the chamber pressure guideline limits, and therefore utilize a reduced (P_{inlet}/P_c) ratio. These cases, therefore, do not reflect the maximum P_c cooling limits.

A rating of enhanced cooling (thermal barriers) for each propellant combination is shown in Table 3 (a rating of 1 represents the best enhancement and 7 represents the least effective enhancement):

Table 3. Rating of Enhanced Cooling

ENHANCEMENT	RATING FOR MAXIMUM P_c		
	O_2/CH_4	O_2/C_3H_8	$O_2/RP-1$
PHYSICAL BARRIER			
CARBON LAYER	5	5	2
ZrO ₂ CERAMIC COATING	3	3	3
GRAPHITE LINER	7	7	7
FLUID BARRIER (3% PERFORMANCE LOSS)			
FILM COOLING	4	4	4
TRANSPIRATION COOLING	2	2	ONA
ZONED COMBUSTION	6	6	6
COMBINED BARRIERS			
ZrO ₂ /FILM COOLING	1	1	-
CARBON/ZrO ₂	-	-	1

Life Sensitivity

A life sensitivity study was conducted to define the influence of the cyclic fatigue life ground rule on maximum chamber pressure. The 600,000 lbf O_2/CH_4 unenhanced chamber design was selected for this study. The life was determined to be related to the chamber pressure by:

$$N_{f_2} = \left(\frac{P_{c_1}}{P_{c_2}} \right)^5 N_{f_1} \quad (24)$$

This relationship should hold true for other thrust levels and propellant combinations using optimum chamber designs. This relationship should be limited to hot-gas wall temperature ≤ 1500 R for copper base alloys. This limit is attained for a life of (100 x 4) cycles and represents a 20% higher chamber pressure than attained for the (250 x 4) cycle life designs.

$O_2/RP-1$ Decomposition Sensitivity

This sensitivity study defined the chamber pressure limits for an unenhanced design and an enhanced carbon layer design as a function of coolant wall temperature. The 600,000 lbf thrust chamber design was utilized for this study. The coolant wall temperature coking limit impacts the chamber pressure level by 600 psi for 100 R change in coolant wall temperature.

Conclusions

Both the unenhanced and enhanced designs reflect an increase in chamber pressure for a decreased

thrust level when cyclic life is the primary fixed criterion. Another contributor to this trend is the shorter length and larger contraction area ratio of the low thrust designs for a 98% energy release efficiency. This chamber pressure to thrust trend is reversed for the $O_2/RP-1$ designs where the coolant wall temperature is the limiting criterion (to meet the coking temperature limit) instead of life limited.

The greatest increase in chamber pressure can be attained by a combined thermal barrier. This combination is a ZrO ceramic coating with film cooling for the O_2/CH_4 and O_2/C_3H_8 propellant combination, and a ZrO₂ ceramic coating with a carbon layer for the $O_2/RP-1$ propellant combination.

The best single heat barrier is transpiration cooling for the O_2/CH_4 and O_2/C_3H_8 propellant combination and a carbon layer for the $O_2/RP-1$ propellant combination.

The highest chamber pressure can be attained for the O_2/CH_4 propellant combination. The $O_2/RP-1$ propellant combination reflects the lowest cooling limits which is primarily attributed to a low RP-1 decomposition temperature. Also, the $O_2/RP-1$ propellant combination must use a high-performance liquid/liquid injector with a short gas/gas chamber design, and depend on a carbon layer to be competitive with the O_2/CH_4 designs.

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